Dr. Jinxin Li

Physicist & Software Engineer

Scientific Computing · C++ / Python · AI Systems

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Professional Profile

Experimental particle physicist and scientific software engineer with nearly seven years of C++ and Python development experience in simulation, numerical analysis, and particle reconstruction. Experienced in technical ownership of scientific software, end-to-end algorithm development, and delivery of validated solutions to an international research collaboration. Also designed and developed PANDA Agent, an AI research assistant combining cross-repository retrieval, evidence-grounded question answering, and systematic quality evaluation.

Technical Skills

Programming & Engineering: C++, Python, Bash, CMake, Git, Linux, Docker.

Scientific Computing & HPC: ROOT, NumPy, Pandas, SLURM, Singularity, Monte Carlo Simulation, Statistical Analysis.

AI & Backend Systems: RAG, LangGraph, FastAPI, REST APIs, PostgreSQL, Qdrant, SQLAlchemy, Hybrid Retrieval, Reranking, Benchmark Evaluation.

Languages: Chinese (Native), English (Proficient, C1), German (CEFR A2).

Professional Experience

Doctoral Researcher / Scientific Software Developer

Ruhr University Bochum - Institute of Experimental Physics I · PANDA Collaboration
Sep. 2019 – Jul. 2026 · Bochum, Germany

- Developed and maintained C++ and Python software for detector simulation, numerical analysis, and particle reconstruction, integrating new functionality across shared scientific codebases.
- Served as Software Owner for the PANDA Luminosity Detector Group's C++ analysis framework, overseeing its maintenance, extension, integration, validation, and final delivery.
- Designed, implemented, and validated particle reconstruction algorithms, delivering research software adopted and used by collaboration researchers.
- Developed detector simulation models and conducted comparative Monte Carlo studies, while supporting software quality through reproducible validation, documentation, and researcher training.

Selected Engineering Projects

PANDA Agent - AI Assistant for Scientific Software & Research

Apr. 2026 – Present · Independent project · github.com/JinxinLee/PANDA_Agent

- Developed an AI research assistant connecting multiple interdependent codebases, scientific publications, and technical documentation to help new researchers and developers onboard faster, understand algorithms, and troubleshoot software issues.
- Designed and implemented Python backend workflows using LangGraph, FastAPI, PostgreSQL, and Qdrant, combining hybrid dense/BM25 retrieval with evidence-grounded question answering and traceable source citations.
- Designed human-reviewed benchmarks and systematic evaluation workflows to guide iterative optimization and assess retrieval quality, answer completeness, and citation reliability.
- Achieved 98.98% Recall@10, 97.18% answer-point coverage, and 100% citation integrity on a development benchmark of 59 questions.

Selected Engineering Projects (continued)

LuminosityFit - C++ Framework Development & Delivery

Mar. 2022 – Jul. 2024 · Doctoral research · github.com/panda-luminosity-detector-group/LuminosityFit

- Extended and maintained LuminosityFit, a C++ numerical fitting framework for luminosity measurement, to handle realistic particle interaction distributions and beam conditions, including numerical algorithm development, debugging, software integration, and validation.
- Completed the framework's final integration and validation, delivering the software for use by the PANDA Collaboration. An end-to-end simulation study demonstrated a (0.05 ± 0.07)% relative deviation from ground truth.

Particle Reconstruction — Algorithm Development & Implementation

Jul. 2024 – Apr. 2026 · Doctoral research · github.com/panda-luminosity-detector-group/RestgasDetermination

- Designed a novel algorithm to reconstruct particle interactions occurring outside the expected collision region, taking the solution from initial concept to practical implementation.
- Implemented the algorithm in C++, integrated it with existing reconstruction software, and validated its performance through simulations, achieving less than 1% relative deviation between reconstructed interaction-density profile and simulation ground truth.
- Delivered the validated algorithm to the PANDA Collaboration, where it was adopted and used by other researchers.

Detector Simulation & Performance Evaluation

Sep. 2019 – Mar. 2022 · Doctoral research

- Designed and implemented a replacement pixel-sensor configuration in a C++ simulation framework, including sensor geometry, physical parameters, and dedicated simulation classes.
- Designed and conducted comprehensive Monte Carlo studies comparing the new design against the baseline across multiple simulation scenarios, particle trajectories, and reconstruction stages.
- Demonstrated up to 50% better spatial resolution, approximately 10% better momentum resolution, and 2.8× the baseline reconstruction efficiency in simulations.

Education

Dr. rer. nat. in Physics

Ruhr University Bochum, Germany · Sep. 2019 – Jul. 2026

B.Sc. in Physics

Nankai University, Tianjin, China · Sep. 2015 – Jun. 2019

Earlier Research Experience

IHEP, Chinese Academy of Sciences · BESIII Collaboration

Research Assistant · 2018 – 2019 · Charm/D-meson analysis and Monte Carlo simulation.

Nankai University · BESIII Collaboration

Undergraduate Research Assistant · 2016 – 2019 · Charmonium analysis and scientific data processing.

AMS-02 Project (CERN / MIT · AMS Collaboration)

Summer Research Assistant · 2017 · Detector simulation and tracker-resolution validation.